

ORIGINAL ARTICLE



Making physical activity fun and accessible to adults with intellectual disabilities: A pilot study of a gamification intervention

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Abstract

Background: Only about 9% of individuals with intellectual disabilities reach the government's physical activity (PA) recommendations. Combining gamification and technology seems particularly promising in overcoming personal and environmental barriers to PA participation.

Method: Eighteen adults with varying levels of intellectual disabilities completed a pilot study to assess the initial effects of a cycling gamification intervention on levels of PA, fitness, psychosocial outcomes, and challenging behaviours. The study comprised three designs: pre-post single group, AB single-case, and qualitative. Social validity, implementation barriers and facilitators were also explored.

Results: Nearly all 18 participants cycled daily. Time and distance cycled daily increase during the intervention while a decrease in stereotyped behaviours was observed. Participants and staff found the intervention enjoyable and socially valid.

Conclusions: Results of the multiple-design study suggest that gamification interventions may be a suitable, enjoyable, and promising way to contribute to PA participation of adults with intellectual disabilities.

KEYWORDS

behaviours, cycling, exercise, game, self-esteem

1 | INTRODUCTION

Participating in organised physical activity (PA) can lead to positive physical, behavioural, social, and psychological outcomes for individuals of all ages and with varying characteristics (WHO, 2021). Within the field of research on individuals with intellectual disabilities, Kapsal et al. (2019) recently conducted a meta-analysis in which they reported that PA has a medium to large effect on a variety of physical [e.g., body mass index (BMI) and muscular strength] and psychosocial

indicators including mental health, well-being, feeling of personal efficiency, and challenging behaviours (e.g., decreasing aggressive behaviours). In their systematic review, Bondár et al. (2020) reported on the psychological benefits of PA for this population, including increased quality of life, self-efficacy, and self-realisation, as well as behaviour changes (e.g., increased levels of PA), but highlighted the low quality of the studies conducted in the field of PA and psychological outcomes of individuals with intellectual disabilities. In parallel, being physically inactive has been associated with multiple negative

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consequences such as health-related complications (e.g., obesity, cardiac problems, diabetes) and poor psychosocial functioning and well-being (e.g., reduced quality of life and social participation) (Center of Disease Control and Prevention, 2019; Di Bartolomeo & Papa, 2016, 2019). Although the benefits of being physically active and the consequences of being physically inactive are well documented, it is estimated that only about 9% (Dairo et al., 2016) of adults with intellectual disabilities reach the government recommendations (i.e., ≥ 150 min of moderate-to-vigorous PA per week; Center of Disease Control and Prevention, 2022).

1.1 | Barriers to physical activity

Many personal and environmental barriers to PA participation by individuals with intellectual disabilities have been documented in the literature. Personal barriers include a lack of intrinsic motivation or interest towards PA (Bossink et al., 2017), a low level of social skills, a high severity of the intellectual disabilities, and the presence of physical disabilities (Alesi, 2017; Bossink et al., 2017; Lucas, 2019). Lack of access to equipment, lack of opportunities to take part in organised activities (Eisenberg et al., 2016), insufficient support by family, peers, or staff, and not having access to transportation (Dixon-Ibarra et al., 2017; Howie et al., 2012) are important environmental barriers that have been repeatedly highlighted in the literature.

Although efforts to make PA accessible to individuals with intellectual disabilities within the community have been put forward (e.g., the development of programmes such as those offered by Special Olympics, n.d.), the above-mentioned personal and environmental barriers can impede on one's ability or opportunity to take part in such activities. The environmental factors seem especially important to consider as many adults with intellectual disabilities are living in specialised housing or spending a large number of hours a week at day centres. Within these environmental barriers (e.g., transportation access and staff having to care for multiple individuals), it is sometimes difficult or impossible for the staff members to accompany individuals to activities offered outside of the setting. Personal barriers, including intrinsic motivation can also make supporting PA in this population challenging for staff members. Moreover, staff often have training related to psychosocial intervention in which they were not necessarily made aware of the importance of being physically active or trained to support individuals in doing so.

1.2 | Gamification and technology for increasing physical activity participation

Over the years, researchers have started developing interventions to increase individuals' (i.e., with or without intellectual disabilities) PA levels, while taking into consideration such barriers. The use of gamification mechanisms and technology seems particularly promising in overcoming many personal and environmental barriers to PA participation, including adults' with intellectual disabilities intrinsic

motivation to be physically active, lack of access to transportation, and lack of opportunities to take part in adapted physical activities (Koivisto & Hamari, 2019). Gamification implies transforming the activity into a playful form (i.e., making in fun). As such, rather than simply participating in a PA, a game is proposed that requires PA expenditure in order to reach objectives or obtain rewards (Mazeas et al., 2022). Gamification can thus contribute to making PA participation intrinsically motivating. As for technology, its use can take many forms ranging from belts and watches that quantify PA to more sophisticated technologies, such as virtual reality tools (see Woessner et al., 2021 for a review of technology-based tools). Due to the progression of technology, it is now possible to integrate technology at low costs in daily PA and in multiple settings (e.g., outdoors and indoors), making such intervention easily accessible.

The number of studies conducted that investigated the feasibility, effectiveness, or efficacy of gamification and technology-driven interventions within the context of PA has proliferated outside of research pertaining to individuals with intellectual disabilities (e.g., to increase PA levels of aging individuals, children, adolescents, adults with chronic diseases; Fogel et al., 2010; Kappen et al., 2019; Mazeas et al., 2022). In contrast, the number and quality of studies implementing gamification and technology components to increase PA is limited among the population of individuals living with intellectual disabilities. However, the results from these studies seem to support their feasibility (possible to implement and practicable) and the potential of such intervention components to increase levels of PA and fitness for this population (e.g., Lancioni et al., 2018; Nabors et al., 2020; Perrot et al., 2021; Ryuh et al., 2022), to lower stereotyped behaviours (Stasolla et al., 2018), and to make PA more enjoyable (e.g., Martinez-Millana et al., 2022; Ryuh et al., 2022; Taylor et al., 2016; Torrado et al., 2022; Yalon-Chamovitz & Weiss, 2008).

1.3 | Gamification and technology driven interventions for individuals with intellectual disabilities

Researchers have created and evaluated different interventions that include gamification and technology components to increase PA of individuals with intellectual disabilities. As an example, Ryuh et al. (2022) tested an interactive dance exergaming intervention using an Xbox 360VR video game console and Kinect accessories with seven adults with mild to moderate intellectual disabilities. The authors found that, after only one 10-min session, participants experienced an increased heart rate, perceived exertion, and reported enjoyment during the intervention. In another recent study, Martinez-Millana et al. (2022) conducted a pilot study with six individuals with mild to moderate intellectual disabilities in which they implemented an interactive cycling intervention that consisted of an exergaming system connected to motion sensor equipment mounted onto the exercise bicycles. The system then projected a cycling video on a tablet or smartphone that allowed the participants to navigate in different virtual environments as they used the equipment. Participants

subjectively described the intervention as enjoyable, and staff stated that they perceived it as an effective means to increase PA participation. In addition, Stasolla et al. (2018) completed a study with five individuals who had Rett syndrome and severe to profound developmental disabilities. Their findings suggest that a technology-based PA intervention can be effective in enhancing participants' mobility, reducing their passivity and isolation, increasing indices of happiness (i.e., mood), and decreasing stereotyped behaviours (Stasolla et al., 2018). In the only randomised controlled trial identified on the subject, Yalon-Chamovitz and Weiss (2008) assessed the feasibility of using a virtual reality intervention to increase PA in adults ($N = 33$) with moderate intellectual disabilities in addition to cerebral palsy. The intervention consisted of a planned 30-min virtual reality session implemented three times a week for 12 weeks. The intervention resulted in enjoyment but did not change participants' self-reported self-esteem. While affordable virtual reality systems exist, the cost associated to purchase one similar to the one used in their study can cost over >10,000\$, making it inaccessible to multiple organisations for individuals with intellectual disabilities.

Although the quality and number of studies are limited, the above findings offer initial support that interventions that include gamification and technology components can be tailored to increase individuals' motivation and enjoyment towards PA. Therefore, such interventions have the potential to offer social significance by leading to changes in individuals' behaviours, while maintaining a high social validity (Langthorne & McGill, 2011). The results also suggest that it is possible to do so directly in a natural setting at a low cost and with minimal support from staff. However, there is a need to obtain a broader overview of the effects of such interventions on different outcomes beyond feasibility and social acceptability, such as levels of PA, behavioural outcomes, and psychosocial outcomes (Ryuh et al., 2022). Moreover, the majority of studies identified in the literature were done with individuals with mild or moderate intellectual disabilities, and most often without other coexisting physical conditions, thus limiting the generalizability of these findings to the broader population with intellectual disabilities. As such, the main objective of the current study was to evaluate the feasibility and effects of a gamification intervention with technological components for individuals with varying levels of intellectual disabilities when implemented in a day centre. Two specific objectives were pursued: (a) measure the short-term (i.e., daily) and medium-term (i.e., after 6 weeks) effects of the intervention on PA (i.e., time and distance cycled), fitness components (i.e., lower body strength and BMI), psychosocial outcomes (i.e., self-realisation and mood), and challenging behaviours; and (b) evaluate the social validity and implementation barriers and facilitators of the intervention as perceived by the participants and staff members. Within the present study, challenging behaviours included behaviours that could physically harm one or others, hinder an individual's learning or interfere with his or her daily functioning (Mahan, 2012; Matson et al., 2010; Myrbakk & Tetzchner, 2008). Based on previous research, it is hypothesised that the intervention would lead to increases in participants' PA, improvements psychosocial outcomes, and lower frequencies and severities of challenging behaviours.

Regarding lower body strength and BMI fitness components, it is hypothesised that positive changes will occur for both measures; however, the size of these changes is expected to be small given the duration of the intervention (i.e., 6 weeks). Moreover, we predict that the proposed intervention will be considered enjoyable, easy to implement, and acceptable by both staff and participants.

2 | METHOD

2.1 | Recruitment

Prior to recruiting the adults with intellectual disabilities, ethical approval was obtained through the first author's institutional research ethics board. To facilitate recruitment, a partnership with a non-profit organisation¹ in the Canadian province of Quebec at which the research project took place was established. Thirty-five adults with intellectual disabilities attend the organisation's day centre, three who had physical conditions making it impossible for them to take part in the study (e.g., quadriplegic, severe arthritis). Staff and the organisation's director were asked to complete the PA Readiness Questionnaire and YOU (Warburton et al., 2018) for each remaining adult to assess the need for a consultation with a medical professional before taking part in a PA intervention (i.e., indoor cycling using a stationary bike). In all, nine participants were screened as having medical condition that required a medical consultation prior to taking part in the study.² No further inclusion or exclusion criteria were used in this study. As such, 23 adults with intellectual disabilities were eligible to take part in this study.

The first author sent an email invitation with a brief description of study to the organisation's director, who then transferred the email to the 23 targeted families.³ Families interested in taking part in the study indicated their availability for an in-person or Zoom meeting with the first or second author. Prior to the meeting, families were also asked to complete the PA Readiness Questionnaire and YOU (Warburton et al., 2018) to assess the need for a consultation with a medical professional before taking part in a PA intervention. Legal consent and adults with intellectual disabilities' assent were obtained for each participant.

2.2 | Participants

Of the 23 families who received an invitation to take part in this study, 18 positively replied. Three families did not return the consent form and two families did not complete the PA Readiness Questionnaire and YOU screening measure prior to the study starting. Finally,

¹To maintain participants' confidentiality, we have refrained from naming the organisation or the town in which it is located.

²Individuals screened as needing a medical consultation were excluded from the study. However, following the approval of a medical professional, they were invited to take part in the gamification intervention, but without completing any of the measures.

³To simplify the manuscript, the use of the term *family* is used to refer to both families and legal representatives.

TABLE 1 Participants' characteristics and information.

Fictitious name	Age	Sex	Ethnicity	Main diagnosis based on DSM	Severity of ID	Other diagnostic or conditions	Comm.	Use of bikes prior to study	Main challenging behavior identified by staff and definition
Eric	24	Male	Caucasian	Autism spectrum disorder	Mod.	ADHD	N-Verbal	Occasional	Stereotyped behaviour: Sequence of behaviours that includes rocking, head movements, and grimacing.
Kevin	45	Female	Asian	Bilateral perisylvian polymicrogyria	Mild	Epilepsy	N-Verbal	No	Crying.
Annie	28	Male	Black	Intellectual disability	Mod.	Generalised anxiety disorder	N-verbal	Occasional	Stereotyped behaviour: Finger movements, rocking, foot and/or hand tapping, vocalisations.
Charles	30	Male	Hispanic	Intellectual disability	Mod.	No	N-verbal	No	Stereotyped behaviour: Grimacing.
Gilles	42	Male	Caucasian	Encephalitis induced intellectual disability	Mod.	No	Verbal	Occasional	Opposition: Non-compliance that includes verbally or physically refusing to respond to request and/or answers questions.
Noah	30	Male	Caucasian	Fragile X syndrome	Mod.	No	Verbal	Regular	Stereotyped behaviour: Repeated pulling on his shirt.
Carole	29	Female	Caucasian	Autism spectrum disorder	Mod.	No	Verbal	Occasional	Stereotyped behaviour: Rocking, arm swaying, finger, finger snapping, foot tapping, face cringing, and screams.
Andy	47	Female	Caucasian	Intellectual disability	Mild	No	Verbal	No	Stereotyped behaviour: Sequence of repetitive hand and finger movements and head swaying.
Sam	32	Male	Caucasian-Hispanic	Down syndrome	Mild	Muscular hypotonia	Verbal	Regular	Disruptive behaviour: Repeatedly interrupting and speaking over others when staff discusses with other participants.
Yvan	49	Male	Caucasian	Microcephaly	Mod.	No	Verbal	Regular	Opposition: Non-compliance that includes verbally or physically refusing

TABLE 1 (Continued)

Fictitious name	Age	Sex	Ethnicity	Main diagnosis based on DSM	Severity of ID	Other diagnostic or conditions	Comm.	Use of bikes prior to study	Main challenging behavior identified by staff and definition
Serge	64	Male	Caucasian	Intellectual disability	Mod.	No	Verbal	Occasional	to respond to request and/or answers questions. Stereotyped behaviour: Grimacing.
Marc	28	Male	Asian	Autism spectrum disorder	Severe	No	N-verbal	Regular	Opposition: Non-compliance that includes verbally or physically refusing to respond to request and/or answers questions.
Cody	71	Male	Caucasian	Intellectual disability	Mild	No	Verbal	Regular	Stereotyped behaviour: Echolalia—repetitive questions.
Justine	42	Female	Caucasian	Intellectual disability	Mild	Schizophrenia	Verbal	No	Stereotyped behaviour: Fixating on an object for at least 3 min).
Molly	41	Male	Caucasian	Intellectual disability	Mod.	Generalised anxiety disorder and ADHD	Verbal	Occasional	Stereotyped behaviour: Repetitive face rubbing.
Tommy	29	Male	Caucasian	Autism spectrum disorder	Mod.	No	N-verbal	No	Stereotyped behaviour: Sequenced grimacing and screams.
Remi	56	Female	Caucasian	Intellectual disability	Mild	No	Verbal	No	Verbal aggression: Confrontational remarks towards others.
Cleo	24	Male	Hispanic	Intellectual disability	Mild	Scoliosis and hypotonia	Verbal	No	Stereotyped behaviour: Pacing.

Note: The information pertaining to participants' diagnosis and severity of intellectual disabilities was extracted for the report that was completed by a qualified professional.

Abbreviations: ADHD, attention deficit disorder with hyperactivity; Comm, communication; DSM, diagnostic and statistical manual of mental disorders; ID, intellectual disabilities; Mod., moderate; N-verbal, non-verbal.

18 adults aged between 24 and 71 ($M = 40.0$, $SD = 14.1$) with intellectual disabilities ranging from mild to severe, participated in this study. The majority of the individuals were male ($n = 13$). Prior to implementing the baseline phase, staff reported that eight participants had never used the cycling equipment, six adults with intellectual disabilities used it occasionally (i.e., approximately once a week, only if solicited by staff), and four regularly (i.e., two or more times a week) in the last 2 months. Two of the 18 individuals had medical conditions (i.e., obesity and cerebral palsy) that made it difficult for them to use

regular stationary bicycles but were able to use hand-activated cycling equipment. See Table 1 for a detailed description of each participant.

2.3 | Staff

Six staff members worked full-time at the day activity centre. Although staff members were not recruited as participants for the gamification intervention, they were asked to contribute to the study

TABLE 2 Staffs' characteristics and background information.

Fictitious name	Age	Sex	Ethnicity	Field of education	Highest diploma	No. of years working at the organisation
Robin	30	Male	Caucasian	Human sciences	Highschool	5.0
Ariel	23	Female	Caucasian	Social work	Cecep	2.0
Caroline	39	Female	Caucasian	Specialised education	Cecep	1.5
Lily	26	Female	Black	Beneficiary attendant	Professional studies	3.0
Grace	18	Female	Caucasian	Human sciences	Highschool	0.5
Nicolas	23	Male	Caucasian	Specialised education	Cecep	1.0

Note: No., number.

(i.e., intervention implementation, data collection, and share their experience in an interview). All six staff members volunteered and consented to contribute to the study. Staff characteristics and background information are presented in Table 2.

2.4 | Gamification intervention

The PA gamification was entitled *Le Tour de France*. Participants were first assigned by staff to teams of four or five. As a team, participants were then invited to work together and accumulated km using the indoor stationary bicycles to progress as far as they could through the 21 stages of the *Tour de France* (<https://www.letour.fr/fr/>) route over an 8-week period.

Different components were added to the environment to make the game motivating and interactive. First, a large map of the *Tour de France* route was printed out and displayed in the room where the stationary bikes were located. Each team's thumb pin was displayed on the map to visualise their progression throughout the route.⁴ Second, a projector and white screen were installed in the room to display a road motion video found on YouTube (e.g., <https://www.youtube.com/watch?v=kfG4pMbUPdw>), giving the participants the impression that they were cycling in the *Tour de France* circuit. Finally, music selected based on participants' preferences and suggestions was played in the background for the participants.

2.5 | Procedure and data collection

Our study spanned over a 10-week period (i.e., June 20, 2022 to August 26, 2022). Each group's daily calendar included a 30-min free play period in the room where the stationary bicycles were installed. During these periods, participants were invited to use the cycling equipment as they wished or could decide to do other activities if preferred (e.g., draw, play a board game). Information regarding when and how measures were taken are presented below.

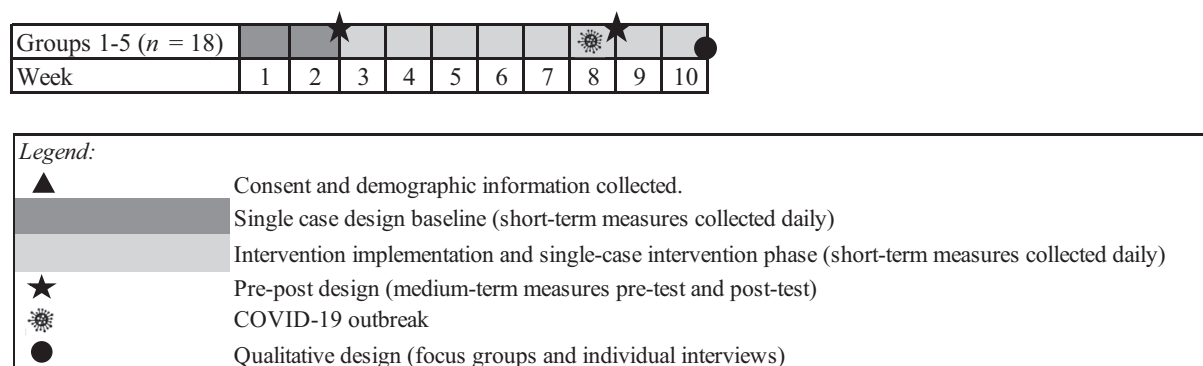
⁴During the first 2 weeks of the intervention, the combined distance cycled by the team would lead to a 1/1 km ratio of progression through the map. However, some participants told the research assistant that they were discouraged as they felt they were not progressing fast. As such, as of week three, we modified the ratio to 10/1 km, meaning that for each km the team cycled, their pin would be moved 10 km on the map.

2.6 | Study design and measures

Given that the intervention was implemented in a natural setting (e.g., the day centre) that offers services to a small number of participants ($n = 35$) and that research tends to support the benefits of PA, it was deemed unethical to conduct a randomised controlled trial or a waitlist trial. Rather, we implemented a variety of complementary designs in order to increase the external validity of the data. Specifically, we implemented a pre-post quasi-experimental single-group design, an AB single-case design, and a qualitative design in the form of focus groups and individual interviews. A pre-post quasi-experimental design "is a quasi-experimental research design in which the same dependent variable[s] [are measured] in one group of participants before (pre-test) and after (post-test) a treatment is administered" (Privitera & Ahlgrim-Delzell, 2018, p. 337). As there is no control group, certain threats to internal validity (e.g., history or maturation) cannot be ruled out when using this design. In contrast to a group design, in a single case design "[...] an individual case [...] serves as its own control with repeated measurements across phases of the study" (Privitera & Ahlgrim-Delzell, 2018, p. 349). In AB single-case designs, there are two phases: the baseline (A) and the intervention (B). Finally, qualitative design "is a type of research that explores and provides deeper insights into real-world problems [...] by collecting data on the participants' experiences, perceptions, and behavior [...] by asking open-ended question on a specific topic or issues" (Tenny et al., 2021, para 1, Introduction). See Figure 1 below for a visual representation of the study timeline of the study procedure.

It is important to note that at week eight of the study, a COVID outbreak in the day centre significantly affected the last 2 weeks of implementation and the post-test data collection. Specifically, three participants and three staff members tested positive for the virus and had to self-isolate for a minimum of 10 days. Moreover, families removed three other participants from the centre during weeks 9 and/or 10 as a preventive measure. As such, following the disclosure of the first COVID case, it was decided by the organisation's director and the research team to immediately proceed to completing the post-test measures for the pre-post quasi-experimental design. This decision was made in order to collect data prior to a potential COVID outbreak in the day activity centre. As such, the post-test

Timeline of the Study Procedure.



Note. Figure 1 shows a visual representation of the study protocol, including the implementation of the three designs and of the intervention, the data collection procedure for the quantitative short- and medium-term measures and qualitative measures.

FIGURE 1 Timeline of the study procedure. This figure shows a visual representation of the study protocol, including the implementation of the three designs and of the gamification intervention, the data collection procedure for the quantitative short- and medium-term measures and qualitative measures.

measure for the group design took place at the end of week 8 and beginning of week 9, rather than as planned at the end of week 10. Regarding the single case data collection, data was pursued as planned; however, the length of the intervention phase varied between eight and 10 weeks as some participants were absent due to COVID during the last 2 weeks of the study. Finally, some of the interviews with the staff members had to be individually conducted using a video conference software rather than through the planned in-person focus groups for three staff members in isolation.

2.6.1 | Single-case design and short-term measures

The AB single-case design was implemented during 10 weeks. The first 2 weeks served as the baseline and the gamification intervention was implemented at week 3 for 8 weeks. Throughout the baseline and intervention phases, a daily 30-min free play period in the room where the stationary bicycles were installed was put in place. At the start of week three, the gamification intervention (i.e., *le Tour de France*, the music, and the cycling videos) was implemented. Within this design, three ad hoc daily measures were included to assess distance and time cycled, the frequency and severity of a behaviour considered as challenging by staff, and mood.

Daily physical activity

Within this study, the type of daily PA measured pertained to participants cycling behaviours. PA outcomes (i.e., time and distance cycled in km) were extracted directly by the cycling equipment and transcribed by a staff member for each participant on every occasion they used the cycling equipment. It is important to note that participants could, upon request, use the cycling equipment outside of the planned 30-min sessions in the room with the cycling equipment.

Daily challenging behaviours

Prior to the study, a meeting with the day centre's staff was conducted to identify a dominant behaviour that was considered challenging by staff for each of the 18 participants (see Table 1 for the behaviour and definition selected for each participant). Challenging behaviour was defined as a behaviour that interferes with the individual's daily functioning (e.g., social interactions, completing an activity). At the end of each day, the primary staff member for each group was asked to rate the frequency and the severity for each participant's selected behaviour using a scale from 0 (*absence or the lowest frequency/severity*) to 10 (*highest frequency/severity of the behaviour*).

Daily mood

At the end of each day, participants were asked by their primary staff member to self-report their mood using a 4-point scale ranging from 1 (*unhappy*) to 4 (*very happy*). Each score was accompanied by a pictogram representing the different intensities of happiness. To increase the reliability of this measure, the scale was implemented in the intervention setting 2 weeks prior to the study starting.

2.6.2 | Group design and medium-term measures

The pre-test post-test single-group design was implemented over a 6-week period. The following outcomes were measured: challenging behaviours frequency and severity, self-realisation, BMI, and lower body muscular strength. The pre-test data collection occurred prior to the implementation of the gamification intervention (i.e., during week 2 of the baseline phase of the single-case design). The post-test measures occurred 6 weeks later.

Challenging behaviours

The main staff member of each participant was asked to report the frequency and severity of the person's challenging behaviours using the French version of the Behaviour Problem Inventory-01 (BPI-01; Rojahn et al., 2001). The BPI-01 comprises 52 items that are divided into three subscales: stereotyped behaviours, self-injurious behaviours, and aggressive/destructive behaviours. The BPI-01 has a good test-retest reliability ($r = 0.76$) and internal consistency reliability (Cronbach alpha = 0.83). Staff were asked to rate the frequency (0 = *never* to 4 = *every hour*) and the severity (1 = *low* to 3 = *severe*) of each individual's challenging behaviours during the last 6 weeks.

Self-realisation

The French version of the 15-item self-realisation subscale of the Arc's Self-Determination Scale (Wehmeyer et al., 2001; Wehmeyer & Kelchner, 1995) was used to measure participants' self-realisation (e.g., "I know what I do best" or "I like myself"). The scale is a self-report measure created for individuals with learning disabilities or intellectual disabilities. Participants are asked if they *agree* (1) or *disagree* (0) with each of the statements. A higher score thus represents a higher level of self-realisation. The subscale was found to have an acceptable internal consistency reliability (Cronbach alpha = 0.64) and acceptable construct reliability, but this value must be interpreted with caution as the sample size used for this analysis was small ($n = 20$ participants with intellectual disabilities). This scale was selected despite its psychometric properties as self-reported measures of self-determination built for individuals with intellectual disabilities are limited.

BMI and muscular strength

The research assistant measured the weight of each participant using the day centre's electronic scale. Height was measured using a 2-m soft measuring tape that was taped to the wall. Participants' BMI (kg/m^2) was then calculated.

Lower body muscular strength was assessed using the 30-s sit-to-stand test that was repeated three times at 5-min intervals. The test consists of recording the number of times a participant comes to a full-standing position in 30 seconds, from a folding chair with a seat height of 17 inches (43 cm), a straight back, and without armrests. Participants' mean score was used for the analyses. This standardised test has undergone different adaptations in various populations and has demonstrated robust psychometric properties (Figueiredo et al., 2021; Zhang et al., 2018). The research assistant was trained by the third author on how to conduct the test.

2.7 | Social validity and intervention barriers and facilitators

During the last week of the study, the second author conducted focus groups or individual interviews with the participants and staff members to assess social validity, barriers, and facilitators of the intervention. The interview guides included semi-structured and open-ended

questions assessing their experiences regarding the intervention and their suggestions for improving the intervention (e.g., making the intervention more enjoyable and sustainable). These questions were created by the research team to extract information pertaining to the social validity which, for the current study, refers "to the social importance and acceptability of treatment goals, procedures, and outcomes" (Foster & Mash, 1999, p. 308). Questions regarding implementation barriers and facilitators of the gamification intervention were also developed (Callahan et al., 2017; Wolf, 1978). As examples, to assess the social validity of the intervention, participants and staff members were asked, "What did you like the most about the tour de France intervention?," "If you could change the Tour de France intervention, what element(s) would you change and why?," and "Did the Tour de France intervention motivate you or discourage you from cycling? Please explain your answer.". Questions pertaining to barriers and facilitators that were discussed with staff members included "What components of the intervention are the most important in your opinion to help pursue its implementation beyond this study?" and "What elements would make maintaining this intervention in the day activity center difficult?" All interviews were recorded using a standard voice recorder.

Focus groups with the participants with intellectual disabilities occurred in groups of two to three participants and pictograms were used to support participants' comprehension of the questions. Moreover, some questions were reframed as close-ended questions (i.e., yes or no) for individuals with lower communication abilities. Interviews were only conducted with verbal participants and those present at week 10 (i.e., not conducted with the six participants absent due to COVID). As such, interviews were conducted with nine participants.

One focus group was conducted with two staff members and individual interviews were completed via Zoom for the three staff members. Two staff members that were in isolation due to COVID-19 could not be reached for the interview.

2.8 | Analysis

Single-case data (i.e., short-term effects) were first organised in graphs. Specifically, the data collected during the baseline and intervention phase for each participants' daily PA, mood, and behaviour identified as challenging by staff were used to create the graphs. The graphs were then interpreted using visual analysis to compare the baseline and intervention data (Lanovaz et al., 2020; Taylor & Lanovaz, 2022). To enhance the reliability of the results, this step was conducted by the first author and an independent researcher who is an experienced behaviour analyst trained in visual analysis of such graphs. The mean interrater agreement was 91.62% ($SD = 6.81\%$, $Min = 82.36\%$, $Max = 100\%$). When accounting for chance, the mean interrater agreement as measured by Cohen's Kappa (κ) was 0.86 ($SD = 0.10$, $Min = 0.74$, $Max = 1.00$). The interrater agreement was considered weak if $[0.40-0.59]$, moderate if $\kappa = [0.60-0.79]$, strong if $\kappa = [0.80-0.90]$, and almost perfect if $\kappa > 0.90$. Disagreements were

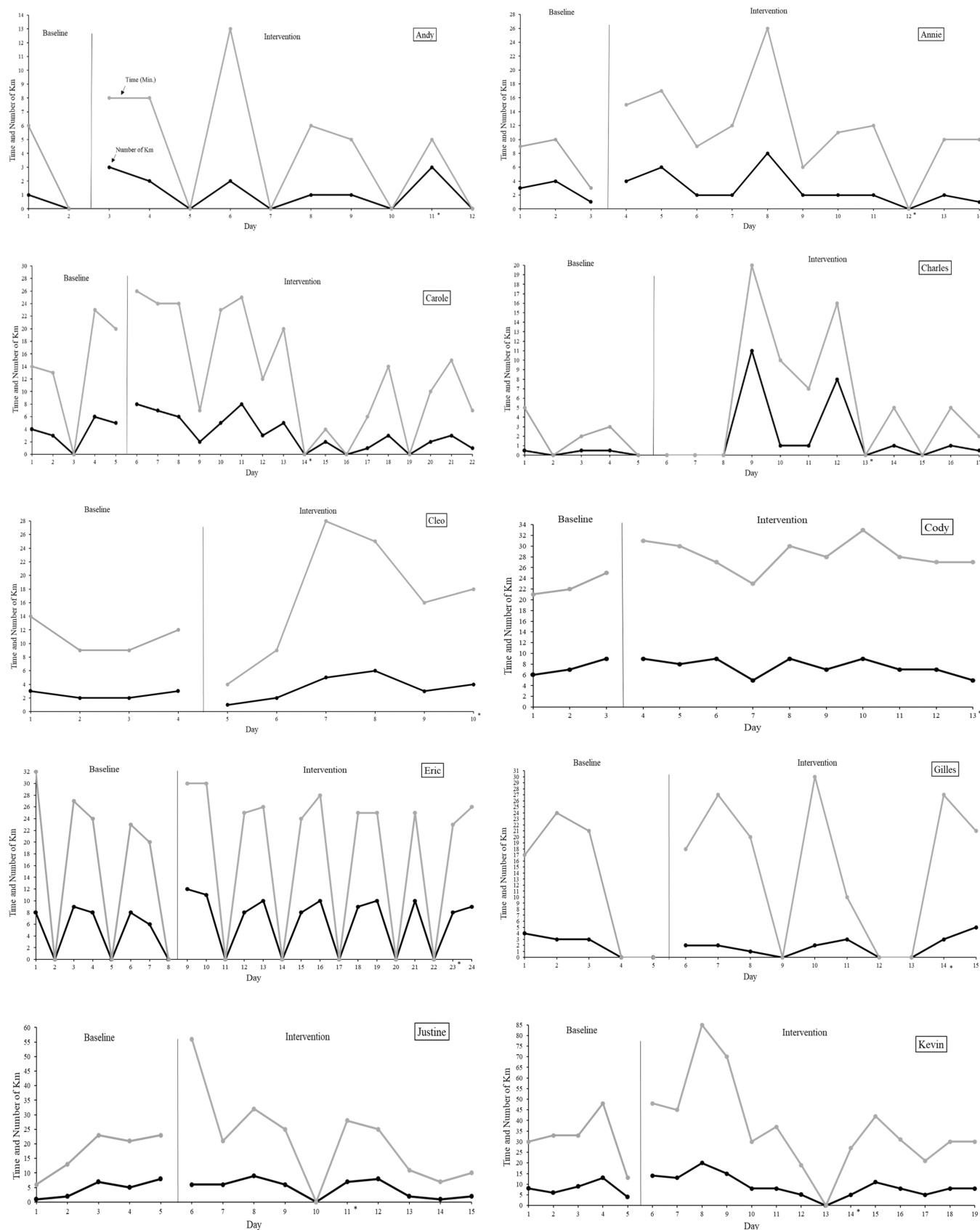


FIGURE 2 Change in time and distance cycled over time. This figure presents the changes in distance and time cycled by the participants. Time is measured in minutes and distance is measured in kilometres. The number of days varies based on the number of days the participants attended the day activity centre during the research project. Andy's results are illustrated in this figure, but were not analysed due to insufficient data during baseline. An asterisk is presented on the x-axis to represent when the Covid outbreak occurred. The interrater agreement was 94.12% and Cohen's Kappa = 0.93 for time and 82.35% and Cohen's Kappa = 0.74 for distance cycled.

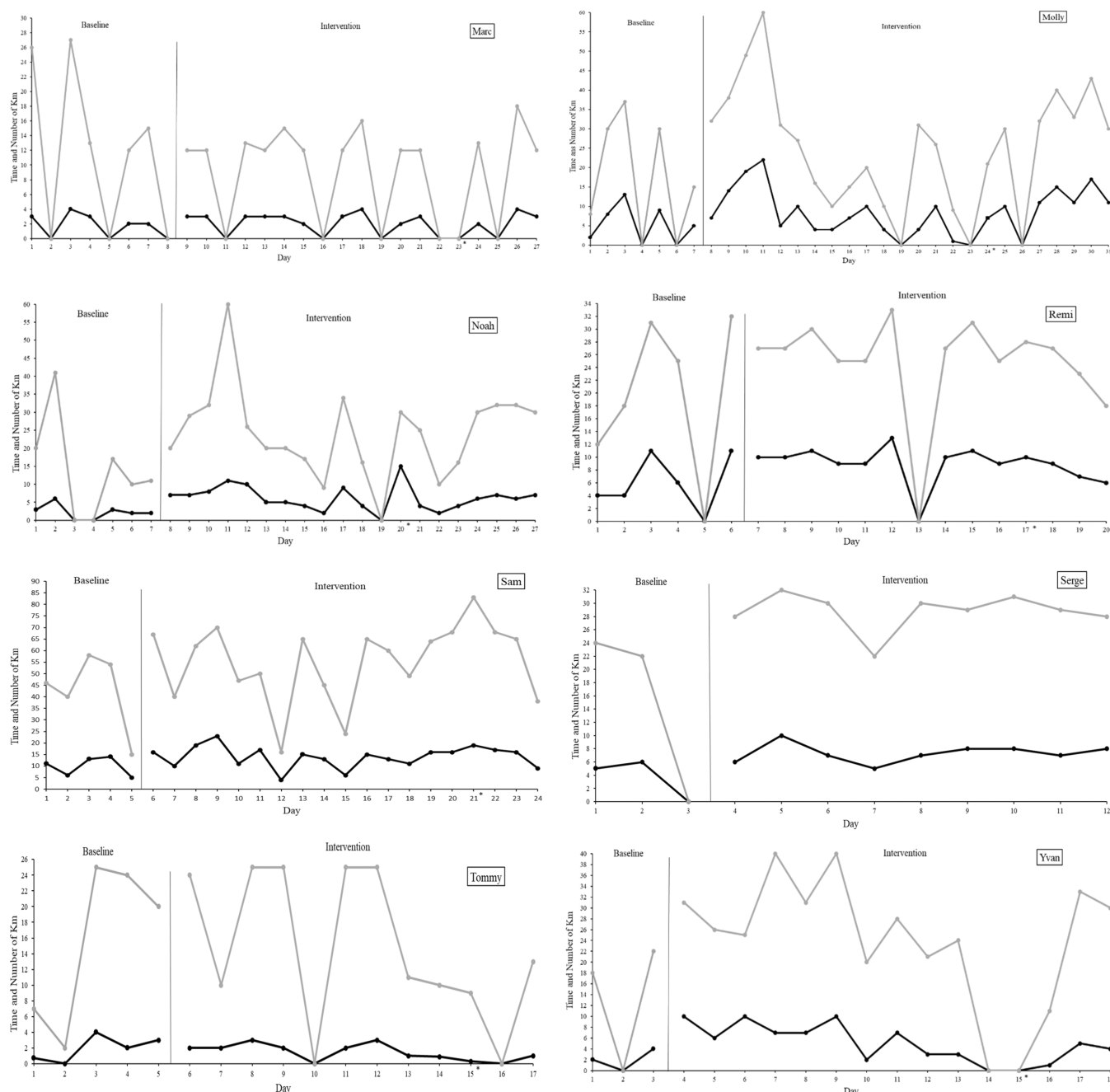


FIGURE 2 (Continued)

resolved by an independent third party (McHugh, 2012). The interrater values for each dependent variable are presented below in section 3. In addition to the visual analysis, descriptive statistics were extracted for the baseline and intervention phases using the short-term data of each participant (see Supporting Information for the descriptive statistics of the short-term outcomes). These data were then used to compare the baseline and intervention phases mean of the group for each of the five short-term outcomes (i.e., daily distance and time cycled, daily challenging behaviour frequency and severity, and daily mood). The nonparametric single-sample Wilcoxon Signed Rank Test (Harris & Hardin, 2013) was used to complete these analyses in IBM® SPSS® version 27 software. This test was selected

rather than a *t*-test as our sample size was small and the data were not normally distributed.

The pre-test and post-test data were also analysed with the non-parametric single-sample Wilcoxon Signed Rank Test (Harris & Hardin, 2013) in IBM® SPSS® version 27 software. As the short-term analyses, this test was selected rather than its parametric analogue given the small sample size and the non-normal distribution of the data. Regarding challenging behaviours, the analysis was conducted for the frequency and severity of each of the three categories of challenging behaviours measured by the BPI-01 (i.e., stereotyped behaviours, self-injurious behaviours, and aggressive/destructive behaviours).

Finally, the qualitative data were first transcribed by the second author. Following the transcription, a deductive thematic analysis was completed to extract the principal themes and responses that emerged from the interviews (Braun & Clarke, 2006, 2019). The thematic analysis was completed using the NVivo® version 12 software. The codes were developed prior to the analysis to extract data relating to social validity as well as implementation barriers and facilitators of the gamification intervention. The social validity codes (i.e., intervention acceptability, outcomes, overall experience) were derived from different definitions of social validity in disability research (e.g., Callahan et al., 2017; Wolf, 1978) as well as inspired by items of the Treatment Acceptability Rating Scale—Revised (Reimers et al., 1991). All citations used in the manuscript were translated to English by the first author.

3 | RESULTS

3.1 | Short-term changes

3.1.1 | Level of physical activity

The interrater agreement for the visual analysis of the AB graphs was moderate ($\kappa = 0.74$) and nearly perfect ($\kappa = 0.93$) for distance and time cycled, respectively. The visual analyses suggest that nine (i.e., Annie, Charles, Cody, Cleo, Molly, Noah, Sam, Serge, and Yvan) and seven participants (i.e., Charles, Molly, Cleo, Noah, Remi, Serge, and Yvan) had a marginal increase in time and distance cycled, respectively. No clear change in time cycled was observed for five participants (Eric, Gilles, Kevin, Marc, and Remi), and eight individuals (Annie, Cody, Eric, Gilles, Justine, Kevin, Marc, and Sam) for distance cycled. Andy had insufficient baseline data (i.e., only two data points) to interpret both PA-level graphs. Although the time and distance cycled varied widely from one participant to another, participants took part in cycling on nearly all baseline and intervention sessions throughout the study. Figure 2 shows the AB single-case graphs with participants cycling behaviours.

When comparing the group's pre-test and post-test cycling behaviours, a significant increase in distance cycled was found ($Ws = -3.02$, $p = .002$, $M_{\text{pre-test}} = 4.26$ km, $M_{\text{post-test}} = 6.74$ km), suggesting that participants cycled an average of 2.5 km per day during the intervention phase. Similar results were observed for time cycled, ($Ws = -3.38$, $p < .001$, $M_{\text{pre-test}} = 15.27$ min., $M_{\text{post-test}} = 19.87$ min), with participants increasing the time they cycled by over 4.50 min a day during the intervention phase.

3.1.2 | Challenging behaviours

The interrater agreement for the visual analysis of the AB graphs was perfect ($\kappa = 1.00$) for challenging behaviour severity and strong

($\kappa = 0.86$) for frequency. Visual analysis of the severity and frequency graphs revealed divergent results between the 18 participants. Specifically, regarding the severity of participants' dominant challenging behaviour, a marginal (i.e., small but detectable) decrease was observed for seven participants (i.e., Carole, Charles, Cleo, Molly, Noah, Sam, and Yvan) while a marginal increase was observed for four other participants (i.e., Cody, Gilles, Justine, and Serge). There was no clear change observed between the baseline and intervention phases for six participants (i.e., Annie, Eric, Kevin, Marc, Remi, and Tommy). Similarly, the frequency of challenging behaviours was found to have marginally decreased for four participants (i.e., Cleo, Eric, Noah, and Yvan) and marginally increase for four individuals (Cody, Justine, Remi, and Serge). No clear change was depicted for nine participants (Annie, Carole, Charles, Gilles, Kevin, Marc, Molly, Sam, and Tommy). Andy had insufficient baseline data (i.e., only two data points) to interpret the AB graph for both the severity and the frequency of his selected challenging behaviour. Of interest, seven of the 11 instances in which a marginally decrease was observed pertained to stereotyped behaviour. Figure 3 presents the AB single-case graphs of each participant for their challenging behaviour.

Regarding the comparison of the baseline and intervention data for the group, no significant changes were found for participants' daily behaviour frequency ($Ws = -1.21$, $p = .227$, $M_{\text{pre-test}} = 3.94$, $M_{\text{post-test}} = 3.43$). Similar results were obtained of participants' daily behaviour severity ($Ws = -0.92$, $p = .356$, $M_{\text{pre-test}} = 2.72$, $M_{\text{post-test}} = 2.42$). These results suggest that, for the behaviours identified by the staff members prior to the study, no significant decreases or increases were observed during the study.

3.1.3 | Mood

The interrater agreement for the visual analysis of the AB graphs depicting participants' mood was on the upper bound of the moderate category with $\kappa = 0.79$. A marginal increase was observed between the baseline and intervention phase for two participants (i.e., Charles, & Marc) self-reported mood, while two participants' (i.e., Cody and Eric) mood seemed to small decrease during the intervention. It is important to highlight that multiple participants reported the highest score on the scale throughout the baseline sessions. Two participants (i.e., Andy and Yvan) had insufficient data in the baseline to interpret the AB graphs for mood. Figure 4 encompasses the AB single-case graphs for participants' mood throughout the study.

The groups' mean daily mood during the baseline and intervention phase were high; with both score above 4.5 on a possible score of 1–5. These results suggest that participants' self-reported mood was generally good throughout the study. The groups' scores during the baseline and intervention phases for this outcome did not differ significantly ($Ws = -1.29$, $p = .196$, $M_{\text{pre-test}} = 4.54$, $M_{\text{post-test}} = 4.74$).

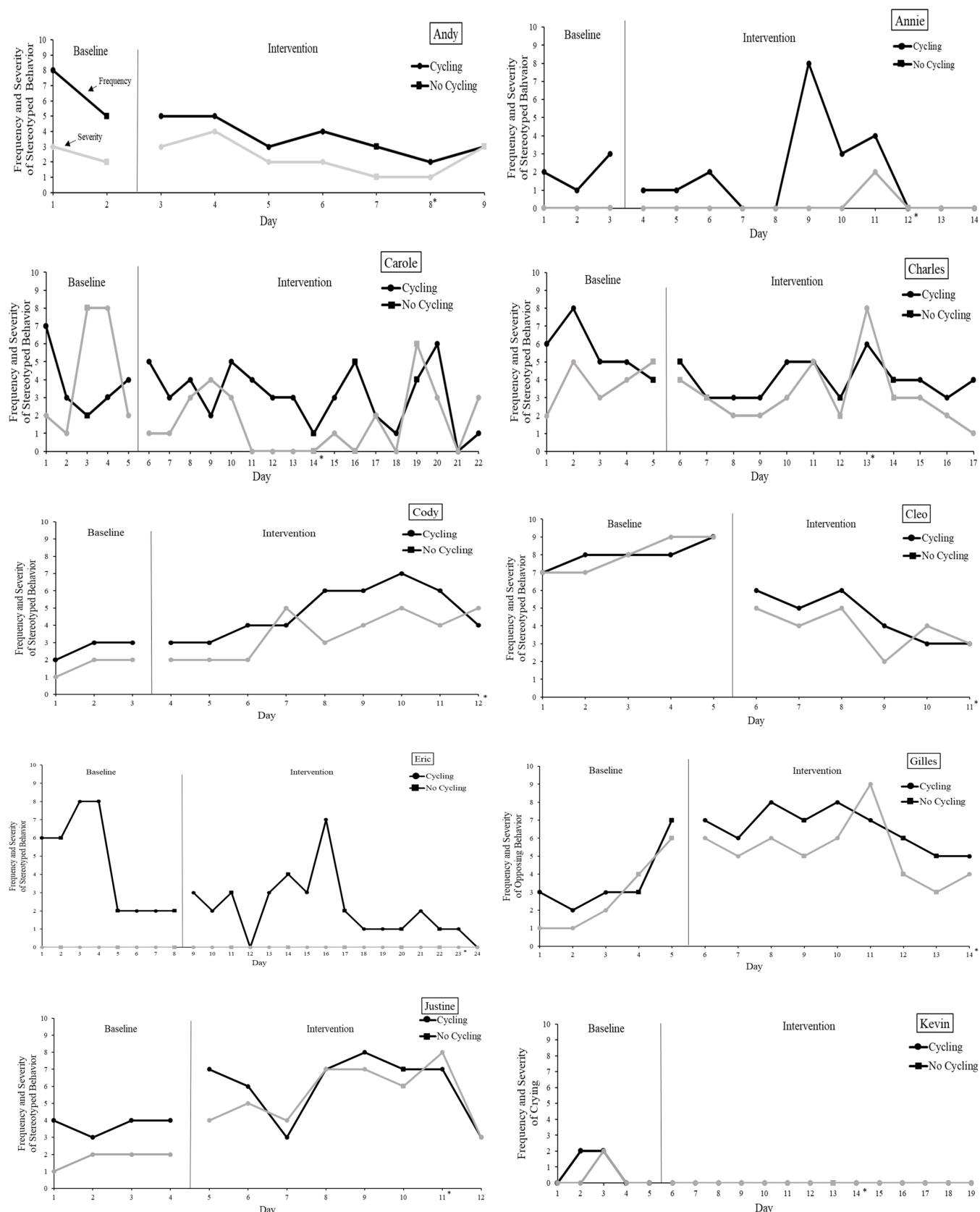


FIGURE 3 Change in challenging behaviour frequency and severity over time. This figure presents the changes in the frequency and severity for each participant's dominant challenging behaviour as measured by staff using a Likert type scale. Lower scores represent lower frequency and severity of the challenging behaviour. The number of days varies based on the number of days the participants attended the day activity centre during the research project. Andy's results are illustrated in this figure, but were not analysed due to insufficient data during baseline. An asterisk is presented on the x-axis to represent when the Covid outbreak occurred. The interrater agreement was 100.00% and Cohen's Kappa = 1.00 for challenging behaviour frequency and 94.12% and Cohen's Kappa = 0.86 for challenging behaviour severity.

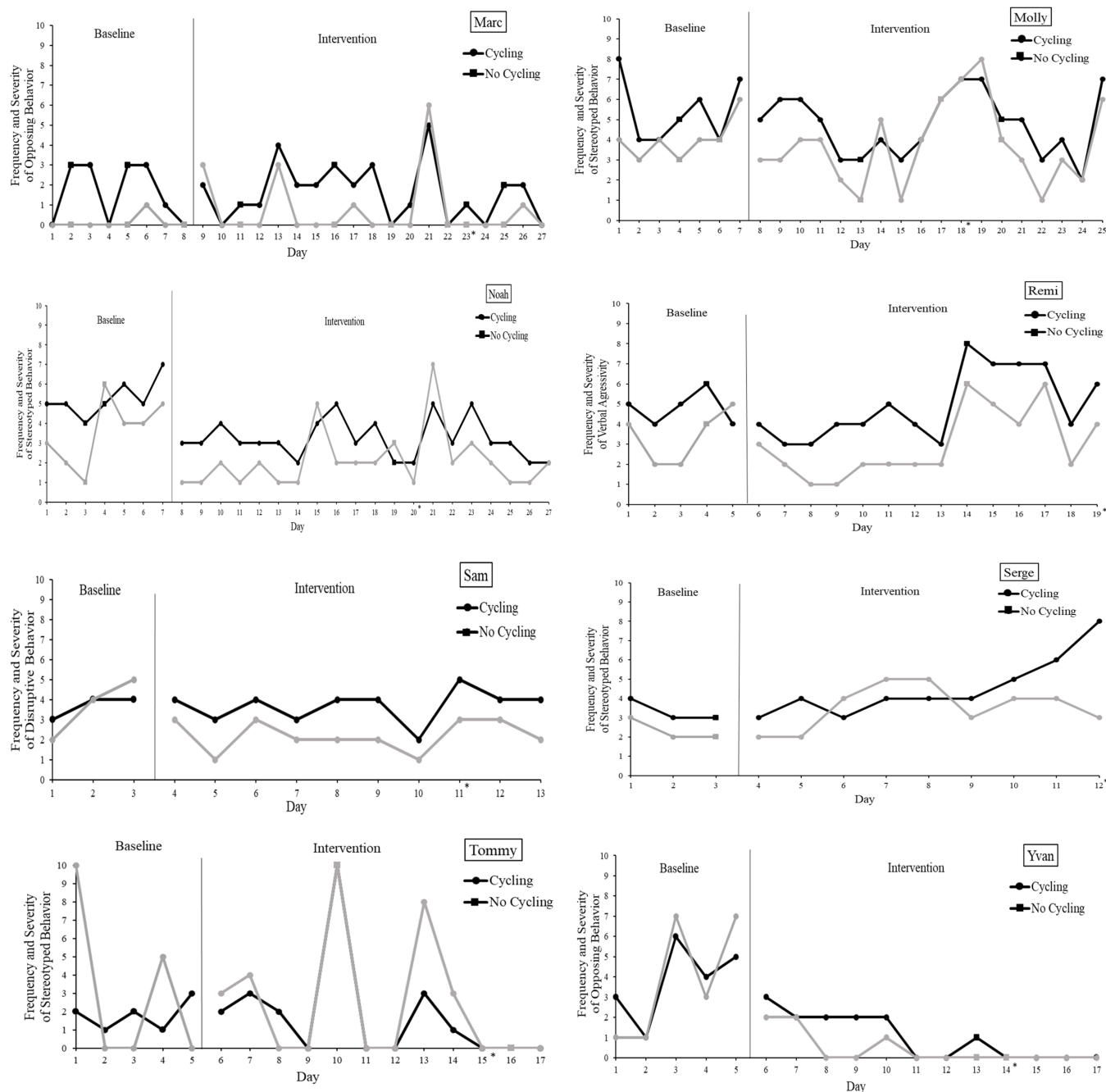


FIGURE 3 (Continued)

3.2 | Medium-term changes

3.2.1 | Challenging behaviours

The BPI-01 was completed at the pre- and post-test for all 18 participants. Regarding the frequency of the behaviours, a significant decrease was obtained for stereotyped behaviours ($W_s = -1.67$, $p = .048$, $M_{\text{pre-test}} = 22.1$, $M_{\text{post-test}} = 19.0$), but not for self-injurious ($W_s = -0.90$, $p = .180$, $M_{\text{pre-test}} = 2.7$, $M_{\text{post-test}} = 2.2$) or aggressive behaviours ($W_s = -0.63$, $p = .264$, $M_{\text{pre-test}} = 2.6$, $M_{\text{post-test}} = 3.3$). In a similar matter, a significant decrease was observed for the severity

of participants' stereotyped behaviours ($W_s = -1.76$, $p = .039$, $M_{\text{pre-test}} = 14.0$, $M_{\text{post-test}} = 11.4$), but not for self-injurious ($W_s = -0.84$, $p = .200$, $M_{\text{pre-test}} = 2.0$, $M_{\text{post-test}} = 1.4$) or aggressive behaviours ($W_s = -0.31$, $p = .379$, $M_{\text{pre-test}} = 2.4$, $M_{\text{post-test}} = 2.8$) severity.

3.2.2 | Self-realisation

In all, 11 participants completed the self-realisation questionnaire at both data collection points. Six participants were non-verbal or their level of comprehension was considered insufficient by staff and family

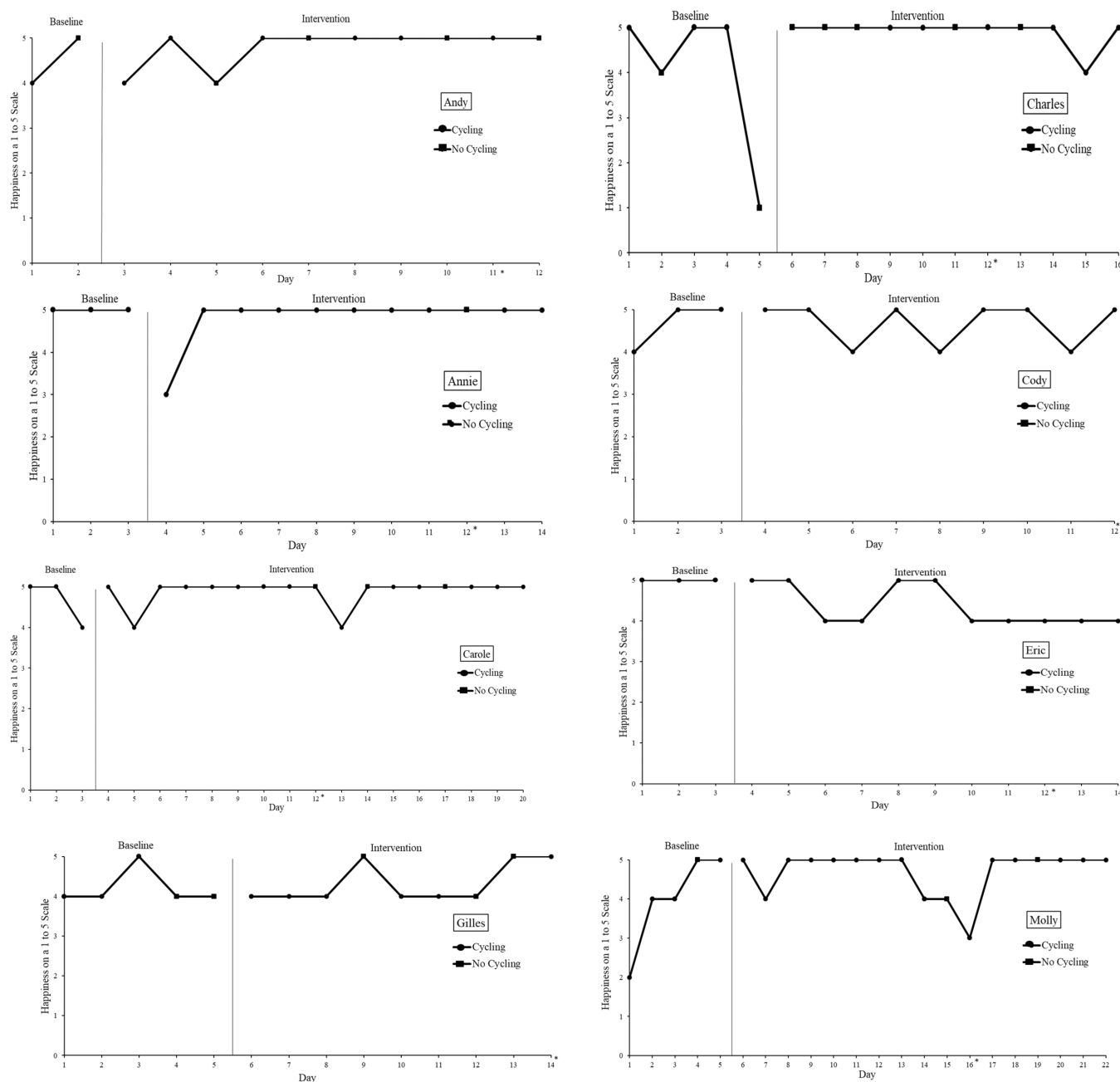


FIGURE 4 Change in mood over time. This figure presents the changes in mood for the participants as self-reported using a 5-point Likert scale. Higher scores represent a more positive mood. The number of days varies based on the number of days the participants attended the day activity centre during the research project. Andy's results are illustrated in this figure but were not analysed due to insufficient data during baseline. Graphs for Remi, Sam, Justine, and Cleo are not presented as participants rated their mood as five during each day of the baseline and intervention (i.e., no variation in scores throughout the study). An asterisk is presented on the x-axis to represent when the Covid outbreak occurred. The interrater agreement was 87.50% and Cohen's Kappa = 0.79 for participants' mood.

members to reliably complete the questionnaire (see Table 1 for participants' characteristics). In addition, post-test data could not be collected for one participant who was absent due to COVID. The Wilcoxon test returned nonsignificant ($W_s = -0.60$, $p = .273$, $M_{pre-test} = 13.9$, $M_{post-test} = 14.3$), suggesting that participants' self-realisation did not change following the implementation of the gamification intervention.

3.2.3 | Morphological outcomes

Due to the COVID outbreak, participants' weight could not be assessed for three participants at the post-test. For the remaining 15 participants, no significant differences were observed for BMI ($W_s = -0.38$, $p = .351$, $M_{pre-test} = 30.5$, $M_{post-test} = 30.6$). Regarding the sit-to-stand test, two participants did not complete the pre- or

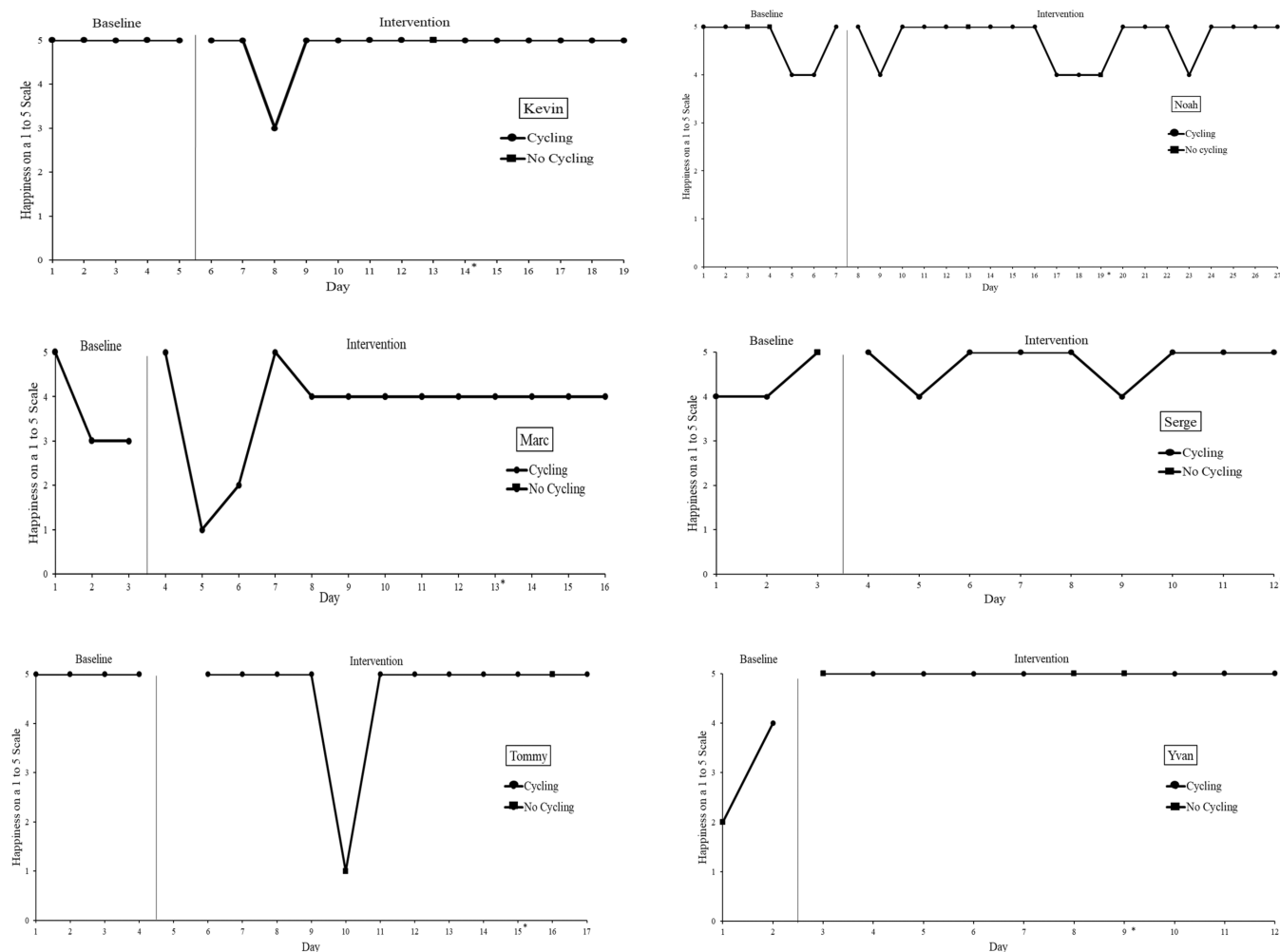


FIGURE 4 (Continued)

post-test measures as they only used the hand-activated cycling equipment. In addition, three participants were not tested at the post-test as they were absent due to COVID. As such, results were analysed for the 13 participants who completed both the pre- and post-test sit-to-stand test. The variation in mean suggests that the number of sit-to-stand repetitions in 30 s, although non-significantly, marginal increased from the pre-test to the post-test ($W_s = -1.60$, $p = .055$, $M_{\text{pre-test}} = 9.5$, $M_{\text{post-test}} = 10.1$).

3.3 | Social validity

3.3.1 | Treatment acceptability and overall experience

During the focus groups and interviews, participants and staff were questioned about their perception regarding treatment acceptability and overall experience. The gamification intervention was regarded as highly acceptable. As stated by Caroline (staff):

It [the gamification intervention] was motivating for the participants, that's what I think increased its value, it gave them motivation with, well, moving forward. Well, basically the most motivating factor is the visual aspect with the projector. I think that it is what led to such a movement of interest, of motivation to cycle because there was a landscape with a road to follow. So, we could really feel their joy to participate.

Participants also reported enjoying the intervention. As an example, Serge said, "I enjoyed cycling, exercising [...] I like pedaling." Only one participant indicated that he did not like the gamification intervention. When asked why, he explained that he wanted to keep everything except for the stationary bicycles as he would have preferred that the intervention occur outdoors "I like it outside, I like to change places" (Yvan).

Staff and participants both reported how being part of a team made the intervention enjoyable and motivating. As stated by Nicolas (staff) "It's as if we were a gang that took part in the Tour de France

together, we talked, we created bonds together, we listened to music.” Cody stated that “It’s fun with the other people,” Serge agreed. Justine said that she did not like to bike alone and that she waited for her friends to use the cycling equipment to join them.

When asked about what they enjoyed most, participants answers varied and included elements such as the music, the map, biking with their friends, the decorations, and having different stationary bikes they could use. Staff consistently reported that they believed that the screen and map were the most appreciated and motivating components of the intervention.

Overall, the interviews show that the intervention was socially acceptable and enjoyable. This is supported by staff and participants’ will to pursue the intervention beyond the research project. Sam even indicated that he wants to tell the organisation’s director that they should travel to Italy next “It is well possible that I will tell the director that we should do a tour from France to Italy next.” In a similar vein, Robin (staff) said, “It was really fun and I hope it will continue.” Finally, Nicolas (staff) indicated that participants want the intervention to continue after the summer break, “Look, they want to continue doing it. So, it’s good for them and for the organization, because it’s something that has been implemented and it’s going to be sustainable.”

3.3.2 | Outcomes

During the interviews, staff members highlighted a couple of changes observed during the implementation of the gamification intervention. The most frequently reported positive outcomes were related to group cohesion, self-esteem, and motivation to be physically active and use the bikes during and beyond the intervention period. Lily (staff) mentioned:

The benefits? Except for those who lost weight thanks to it, that’s already good honestly. Some of them arrived stressed [at the center], I don’t know, in a state... And after they cycled, I could see that they were very calm and that things were doing very well. So, for me, there were lots of benefits.

Participants also reported how the intervention made them feel good. Carole explained that “It [cycling] is good for my body, it’s to lose weight” and later added that “Before I didn’t like riding a bike, now I like it.” Similarly, Cody indicated that “[cycling] is a great way to work out your legs and my belly.” When asked how the game made him feel, Serge responded, “proud of myself.”

No staff member reported changes in behaviours considered as challenging, but some stated that the intervention seemed to enhance participants’ overall mood. The only negative outcome of the intervention that was reported by staff was related to having to intervene when participants wanted to use the same bike: “If they didn’t have their bike, they didn’t want to cycle, but that’s how much they liked it and that’s the good thing, that they liked it” (Robin—staff).

3.3.3 | Implementation barriers and facilitators

Staff members said that the overall implementation of the intervention went well, in large part due to the intrinsic motivation of the staff and director: “Well it worked well, so there aren’t that many barriers, but in other community organizations, if the director is not supportive, if the team is not organized, if they are less conscientious, it may be difficult to carry out.” (Robin—staff). Creating a schedule for the cycling activity was viewed as a facilitator by some staff members but as a barrier by others: “It was definitely more organization for me because we had to respect the schedule, and if we didn’t respect the schedule we couldn’t go and do it later, because the gym was taken after.” (Robin—staff).

The majority of the other barriers extracted from the interviews were related to environmental factors. The most frequently reported barrier was the time necessary to collect data for the *Tour de France* game (i.e., time and distance cycled for each participant), combine the teams’ data, and manually move the thumb pins on the map. Lily (staff) mentioned that “I understood the grids right away, it wasn’t really a problem for me, but it was more about the time I had to do all that, that was the problem in fact.” To resolve this problem, Nicolas (staff) suggested using technology in the future such as sensors that can automatically extract such data: “it’s sure that it would be even better if it [the data] was automatically compiled by technology.” Finally, lack of training and comfort to use the technology (i.e., the laptop and projector) were other elements reported as implementation barriers: “There are many [staff members] who are comfortable with the computer, the projector, but some had no knowledge on the cycling equipment and that was more difficult [...] The training would be, I believe in this area.” (Caroline—staff)

4 | DISCUSSION

The general goal of the present study was to pilot test a low-cost gamification intervention to increase PA levels of adults with intellectual disabilities when implemented in a natural setting. Two specific goals of the research project were to (a) measure the short-term (i.e., daily) and medium-term (i.e., after 6 weeks) effect of the intervention on PA, fitness components, psychosocial outcomes, and challenging behaviours; and (b) evaluate the social validity, barriers, and facilitators to the implantation of the intervention as perceived by the participants and staff members.

The results suggest that most participants took part in the cycling activity daily throughout the study and enjoyed the intervention. Moreover, the cycling activity seemed to increase group cohesion in the day activity centre and was considered easy to integrate into the participants’ routine by staff. In addition, the visual analysis of the AB single-case graphs suggests that participants’ PA levels (i.e., time and distance cycled) marginally increased for approximately half of the adults with intellectual disabilities. For the majority of the remaining participants, their PA levels were maintained during the intervention period. In addition, the group’s average time and distance cycled daily

were found to significantly increase during the intervention phase. Although these changes are marginal in size (e.g., an increase of about 4.5 min of cycling per day), these results suggest that a gamification intervention is promising to encourage and maintain PA engagement. Future studies need to assess components that could help increase the size of this change and thus help adults with intellectual disabilities meet the daily physical activity guidelines. In addition to changes between the baseline and intervention phases, perhaps the most important finding of this study is the change in PA participation that occurred following the implementation of a planned PA period in participants' daily schedule during the baseline period. Specifically, prior to the start of the study (i.e., before the baseline phase), the majority of participants did not use the cycling equipment that was available in the day centre or only used it occasionally (see Table 1). In order to make data collection easier for the staff, it was decided that each group would plan a 30-min free play period in the room where the cycling equipment was present. This small change in the organisation at the start of the study (i.e., during the baseline phase) resulted in an increased use (i.e., from no use or occasional use to daily use) of the cycling equipment by 14 participants. As seen in previous studies, incorporating planned free play periods in interventions for individuals with intellectual disabilities can contribute to their physical and social development (McGarty et al., 2021).

Although nonsignificant, a positive trend in participants' muscular strength was observed for participants who took part in the pre- and post-test measures. While encouraging, two important limitations surrounding this outcome must be noted. First, many participants started using the cycling equipment daily during the 2-week baseline (i.e., prior to implementing the gamification and technology components). As such, this may have influenced the pre-test results. Second, the duration of the intervention implementation was short. Although previous findings have suggested that PA interventions with similar durations can lead to some measurable physical changes (see Bartlo & Klein, 2011 or Obrušnikova et al., 2022 for a systematic review and meta-analysis), it is possible that the intensity at which the participants cycled was insufficient to lead a significant change during the intervention phase. Replication of this study while controlling for these limitations is important to better understand the potential fitness benefits of PA gamification interventions.

Regarding challenging behaviours, when combining the data collected for the behaviours selected by the staff for each participant, no significant changes in daily challenging behaviour frequency and severity were observed. In contrast, the visual analyses of the short-term data and the results of the medium-term analyses suggest that some changes occurred during the intervention. More specifically, these changes generally pertained to behaviours that can be classified as stereotyped or self-stimulatory. This is coherent with other studies that have found that PA can result in decreased self-stimulatory behaviours in populations including those with intellectual disabilities or on the autism spectrum (see Borland et al., 2022; Tarr et al., 2020 for reviews). A plausible explanation is that PA, if done at a certain intensity (e.g., moderate or vigorous) can result in the secretion of neurotransmitters (e.g., serotonin, dopamine; Ma, 2008) that, when

not sufficiently secreted in an individual, have been linked to the adoption of self-stimulatory behaviours (e.g., Keller et al., 2021; Kim et al., 2016). It is important to note that stereotyped behaviours are not unanimously considered as "challenging behaviors." A number of researchers and advocates support the premise that such behaviours serve to regulate or cope with stress and anxiety (see Williams et al., 2021 for a systematic review on this topic). Within this study, staff could identify stereotyped behaviours as challenging behaviours given that, even though they may serve such an adaptive function, they can also impede one's capacity to fully participate in certain environments (e.g., the day centre). We warrant that it would be unethical for interventions to aim to suppress stereotyped behaviours with the assumption that they are solely "problematic" or "challenging." The PA intervention proposed in this study did not intend to decrease stereotyped behaviours, but rather to measure changes in behaviours that can be considered as challenging that occurred during and after this intervention. On the other hand, the intervention did not seem to be associated with short- or medium-term changes for other types of behaviours including aggressive behaviours, oppositional, and self-injurious behaviours. It is important to highlight that some participants' challenging behaviours increased during or after the intervention. As an example, the single-case data suggested a marginal increase in the frequency or severity of the participant's dominant challenging behaviour for Cody, Serge, Justine, Remi, and Gilles. When comparing the graphs with the PA data (i.e., the distance and time cycled) and the severity and intensity of the dominant challenging behaviour graphs there did not seem to be a relation between these variables. In addition, the interviews conducted with these five participants suggested that they enjoyed the intervention. As such, it is possible that other external factors that were not measured during this study (e.g., sleep patterns, change in medication, weather) may help explain these changes.

The qualitative and quantitative results suggest that participants' mood was positive on the days they cycled. However, no significant increases were detected in participants' mood between the baseline and intervention phases. The presence of a ceiling effect (i.e., most participants reporting the highest score on the scale throughout the baseline phase) limited the detection of such an effect. Similarly, participants' self-realisation scores did not significantly increase between the pre- and post-test data collections. As per the self-reported measure of mood, it is plausible to consider that participants' high scores at the pre-test data collection (i.e., $M = 13.9$ out of a possible 15) may have affected the measure sensitivity in detecting change. In addition, the results of the self-realisation subscale must be interpreted with caution as the psychometric properties of the scale were found to be of poor quality (Wehmeyer et al., 2001).

Data from the focus groups and individual interview support that the gamification intervention was regarded by staff and participants as socially acceptable, fun, generally easy to implement, and something that could and should be maintained beyond the research project to help individuals with intellectual disabilities be physically active. Contrary to Mazeas et al.'s (2022) who indicated that having intellectual disabilities prevents "[individuals] from engaging in PA or

understanding the principles of the game" (p. 3), this study, in addition to other recent studies (e.g., Lancioni et al., 2018; Martinez-Millana et al., 2022; Stasolla et al., 2018) adds to the understanding that individuals with intellectual disabilities can actively take part in and appreciate gamification interventions. Research has suggested that individuals with intellectual disabilities seem to have low intrinsic motivation towards PA (Farrell et al., 2004; Ryuh et al., 2022); thus, gamification interventions can potentially serve to enhance intrinsic motivation by increase enjoyment of being active (Jozkowski & Cermak, 2020). Nevertheless, one important element highlighted by a number of staff members was the need to simplify the data collection procedure related to the implementation of the game. Simplifying the data collection procedure could serve to enhance the social validity of the gamification intervention and potentially increase implementation reliability and adherence in natural settings (i.e., outside of a research setting). However, many affordable apps that can automatically collect such data need to be connected to a cadence or speed sensor as well as a power metre (i.e., a tool to measure the power that is output by the person cycling). Such technology is not always integrated into standard cycling equipment or the cycling equipment does not have a Bluetooth setting to connect it to an app. As an example, none of the stationary bicycles purchased by the centre had wheels on which we could attach a cadence or speed sensor. Researchers need to identify potential solutions to simplify or automate data collection when using cycling equipment in gamification interventions.

4.1 | Study limitations and future directions

Although the results from this pilot study are encouraging, a number of limitations restrict the conclusions and generalisations that can be drawn. First, the PA gamification intervention was only implemented during a period of 6–8 weeks. Although research suggests that gamification and exergaming interventions can be promising in increasing motivation and adherence to PA, it is important to evaluate their long-term effectiveness on such variables as most of the interventions tested are short-term interventions (e.g., 12 weeks or less). Moreover, although we measured PA indicators, our results do not allow us to conclude if participants met PA guidelines as we did not measure the intensity (i.e., light, moderate, vigorous) of the cycling activity by the different participants. Including such measures is important to better understand the potential benefits of gamification interventions as some studies suggest that varying levels of PA may result in different psychosocial and physical outcomes (e.g., Aylett et al., 2018; Center of Disease Control and Prevention, 2022; Costigan et al., 2019). However, knowledge regarding PA dosage/intensity and outcomes is lacking (Bull et al., 2020). Thus, practitioners and managers of facilities and organisations serving people with intellectual disabilities should establish daily schedules aligned with evidence-based guidelines. For instance, adhering to the 24-h movement guidelines is recommended, which includes limiting daily screen time (i.e., reducing sedentary behaviour), promoting good sleep hygiene, and ensuring 1 h of daily

PA to contribute positively to the mental health of this population (Canadian Society for Exercise Physiology, 2021). Finally, the use of multiple research designs (single-case design, group design, and interviews) within our study is considered a strength; however, replication of our findings with a larger sample and using a randomised controlled trial is necessary to contribute to the validity and reliability of data on gamification interventions to increase PA participation of individuals with intellectual disabilities. The implementation of a cluster randomised waitlist trial with multiple organisations or day activity centres, where the randomisation occurs by centre could be an ethical avenue. Further research is needed on intervention strategies to increase PA levels of individuals with intellectual disabilities, but efforts should be put forward to be as inclusive as possible (i.e., create interventions for individuals with all levels of intellectual disabilities with and without co-existing psychological or physical conditions).

4.2 | Conclusion

Together, the quantitative and qualitative results of our study point in the direction that gamification interventions may be a suitable, acceptable, and effective way to contribute to PA participation of adults with varying levels of intellectual disabilities. The implementation of gamification interventions that require minimal equipment in a setting such as a day centre offers multiple advantages including not requiring transportation, low cost, and adaptable to participants with varying cognitive and physical conditions. Furthermore, the intervention provides an encouraging avenue to enhance the health benefits of individuals with intellectual disabilities through their participation in PA. Gamification interventions to increase PA levels seem promising for a variety of populations, but further research is needed to better understand essential components and the underlying mechanisms of such interventions.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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